

Johnathan Aguilar

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Education

University of California, Los Angeles

Sept 2026 – Expected 2028

M.S. in Computer Science

- GPA: --/4.0

California State University, Northridge

Aug 2023 – May 2026

B.S. in Computer Science, Minor in Data Science

- GPA: 3.95/4.0

Research Experience

Research Intern

Pasadena, CA

NASA Jet Propulsion Laboratory (JPL)

June 2026 – Present

- Developed YOLO and CNN-Transformer machine learning pipelines for pattern recognition in large-scale satellite imagery.
- Boosted model recall by 33.8% and $F_{0.5}$ score by 20.1% by engineering domain-informed preprocessing and multimodal features.
- Built an end-to-end ML workflow, from data annotation and spatially independent dataset splitting to threshold optimization and evaluation.

Undergraduate Researcher

Northridge, CA

California State University, Northridge - Department of Computer Science

Oct 2025 – May 2026

- Processed 10 years of daily NASA/JPL MUR Level-4 SST data by building a comprehensive remote-sensing analysis pipeline.
- Identified dominant spatial patterns and clustered temporal behaviors using spatio-temporal ML methods like EOF decomposition, DTW, and K-means.
- Designed visual analytics, including heatmaps, Hovmoller diagrams, and cluster maps, to interpret high-dimensional geospatial imagery.

AI Research Intern

Sun Valley, CA

Waste Management & Autonomy Research Center For STEAHM

Sept 2025 – May 2026

- Fine-tuned vision-language models (GPT-4o, Gemini, Qwen) for handwritten table recognition on just 90 samples, achieving 98.6% TEDS accuracy against six OCR baselines.
- Engineered a multimodal RAG agent using LangGraph, CLIP, and FAISS to dynamically route text and image queries for evidence-grounded responses.
- Built an automated Ragas and LLM-judge evaluation pipeline benchmarking Text-KG RAG against baseline RAG across 121 queries for faithfulness, answer relevancy, context precision, correctness, and completeness.

Publications

KEMMA: A Knowledge-Enhanced Multimodal Multi-Agent Framework for Training, Analysis, and Decision Support in Smart Manufacturing,

Target journal: Journal of Manufacturing Systems, manuscript in preparation (2026)

Xuan Liu, **Johnathan Aguilar**, Shangyuan Liu, Hossein Khandandel, Kevin Vaughn, Mark Grady, Jerry Ying Hsi Fuh, Wen Feng Lu, Eugene Tseng, Nhut Ho, Bingbing Li

Preprocessing and Deep Learning Strategies for Internal Solitary Wave Detection in SWOT Observations,

Target venue: IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS), manuscript in preparation (2026)

Johnathan Aguilar, Mathew Archer, Joe T. Roberts

Presentation & Posters

Statistical Analysis of MUR Sea Surface Temperature: Approaches to Thermal Refugia Detection in the Seafloor MPA

Feb 2026

Lightning talk presentation at the LA Geospatial Summit, University of Southern California.

Authors: **Johnathan Aguilar**